time

Service Description

Service ID: *"time"*

Abstract

This document describes the `time` service, version 1.0.0, produced by the TimeManager support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `time` service, which provides the current time of the local cloud with the configured time zone and an assessment whether the time can be trusted (offset to NTP below a threshold). It is produced by the TimeManager system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

Successor of the Arrowhead 4.6 TimeManager `time` service; the WebSocket push and `/echo` were removed.

2.2 How This Service Is Meant to Be Used

Systems that need a common or trusted time call `get-time` and check `trusted` before relying on it.

2.3 Important Delimitations and Dependencies

Trust is based on SNTP checks against `ntp.server.list`; the time is untrusted until the first check and while no server answers. With `service.authorization.enabled=true` every call is checked with Consumer Authorization.

3 Service Interface

This section describes the interfaces to the `time` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

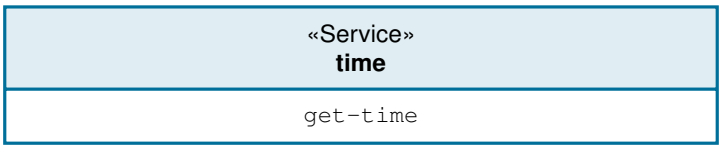


Figure 1: SysML-style block of the `time` service and its operations.

The following interface operations are available.

3.1 operation get-time () : TimeResponse

Returns the current time, time zone, DST flag and trust assessment.

4 Information Model

Here, all data objects that can be part of the `time` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.2, which are used to represent things like hashes and identifiers.

4.1 struct TimeResponse

Current time and its trust assessment.

Field	Type	Description
epoch	Number	Current time, seconds since 1970-01-01T00:00:00Z.
epochMs	Number	Current time, milliseconds since 1970-01-01T00:00:00Z.
tz	String	Configured IANA time zone (`time.timezone`).
dst	Boolean	Whether daylight saving time is in effect in that zone.
trusted	Boolean	Whether the clock offset to NTP is below `time.offset.threshold`.
offsetMs	Number	Last measured offset to the NTP server in milliseconds; null before the first check.
timeSource	String	NTP server of the last successful check.
lastCheckTime	DateTime	Time of the last successful NTP check.

4.2 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “TimeManager – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-TimeManager-java-spring/documentation/v5.3.0/TimeManager-v5.3.0-SysD.pdf.

[2] ——, “time HTTP/TLS/JSON – Interface Design Description (TimeManager),” Eclipse Arrowhead, IDD 5.3.0, 2026, timeManager-v5.3.0-time-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	